

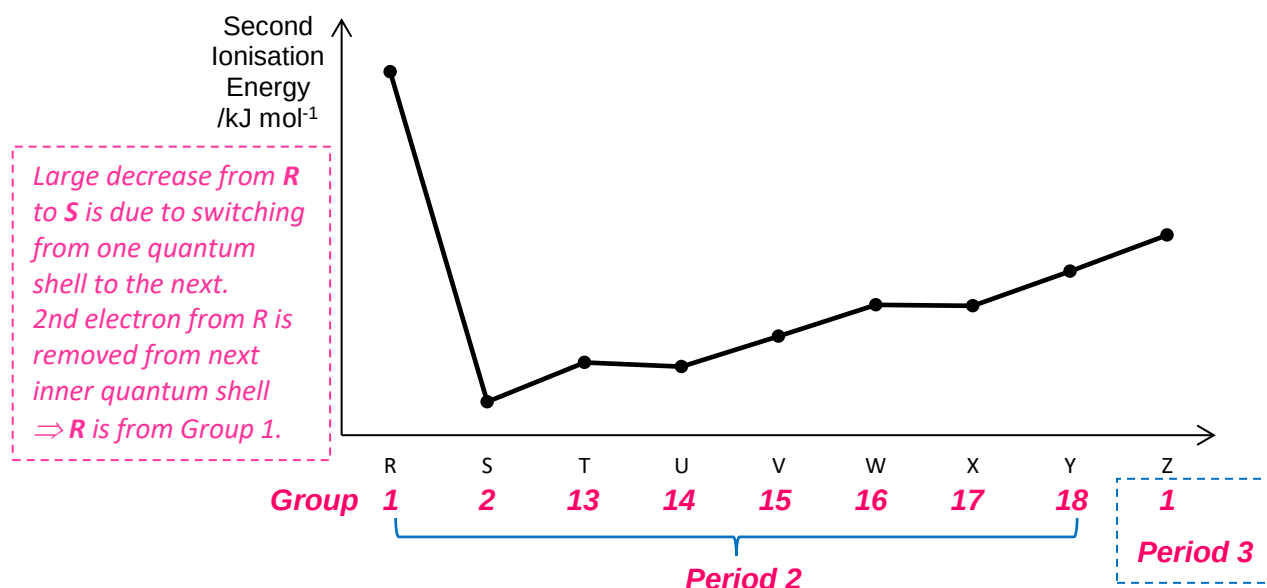
2021 TMJC Preliminary Examination

H1 Chemistry Paper 2 (Answers with Markers' Feedback)

Section A

- 1 Ionisation energies can be measured by finding the minimal energy required to remove the least bound atomic electrons in an evacuated tube.

- (a) Fig 1.1 shows the variation of the second ionisation energies for 9 consecutive elements R to Z with atomic numbers below 18.



- (i) Explain the difference in the second ionisation energy between

1. elements V and W

- From V to W, nuclear charge increases but shielding effect remains relatively constant. [1]
- As effective nuclear charge increases, electrostatic forces of attraction between nucleus and valence electron increases. Hence, greater amount of energy is required to remove the valence electrons from W⁺ and the 2nd I.E. of W is greater than V. [1]

2. elements W and X



There is inter-electron repulsion between paired electrons in valence p orbital of X⁺ (or in doubly-filled valence p orbital of X⁺). Hence, lower amount of energy [1] is needed to remove the valence electron from X⁺ than W⁺, and the 2nd I.E. of X is lower than W.



3. elements R and Z

Z is directly below R in periodic table, i.e. same group.

- Down the group, despite the increase in nuclear charge, there is an increase in the number of occupied electron shells and shielding effect. [1]
- The valence electrons are less strongly attracted to the nucleus, and less energy [1] is required to remove the valence electron of Z^+ . Hence, the 2nd I.E. of Z is much lower than R.

[5]

Marker's Comments

- For 2nd I.E., electrons are being removed from cations with +1 charge, i.e. W^+ and X^+ , not atoms.
- V vs W: generally well-explained. Note that effective nuclear charge $\neq$ nuclear charge.
- W vs X: Many have forgotten the specific phrases (e.g. paired electrons in valence p orbital of X^+) necessary for credit to be given.
- R vs Z: Many did not realise that this question involves a 'down the group' I.E. trend explanation.

(ii) State the Group number for element S.

Element S is from Group 2 [1] (no reasoning required)

[1]

- (b) Element Q also has atomic number below 18. Fig. 1.2 shows its first eight ionisation energies.

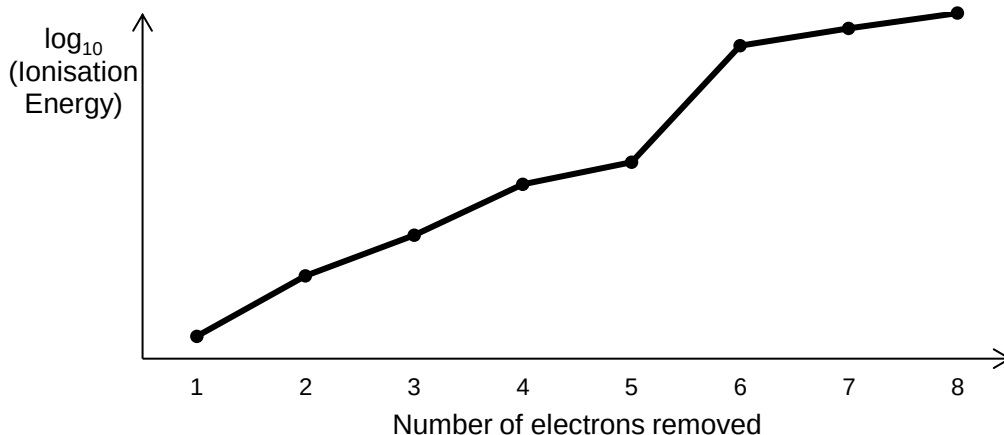


Fig. 1.2

(i) Deduce the Group which element Q belongs to.

- There is a sharp increase between 5th and 6th ionisation energy.
- The 6th electron is removed from an inner quantum shell which experiences much stronger electrostatic force of attraction to the nucleus.
- Only the first 5 electrons are in valence shell.
- Element S is from Group 15.

Any 2 points – [1]; All 4 points – [2]

[2]

(ii) Hence, write the electronic configuration of element Q.

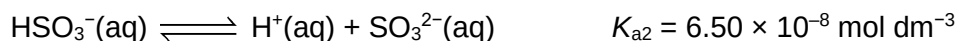
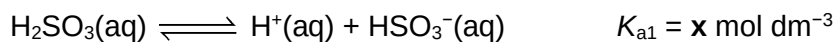
$1s^2 2s^2 2p^6 3s^2 3p^3$ [1]

[1]

[Total: 9]



- 2 Sulfur dioxide dissolves in water to form a weak dibasic acid, H_2SO_3 , which dissociates as follows:



30.0 cm^3 of a 0.20 mol dm^{-3} solution of H_2SO_3 was titrated against a solution of 0.30 mol dm^{-3} KOH. The neutralisation process of a dibasic acid proceeds stepwise, where each equivalence point corresponds to the neutralisation of H^+ in K_{a1} and K_{a2} respectively. The resulting pH curve is shown in Fig. 2.1 below.

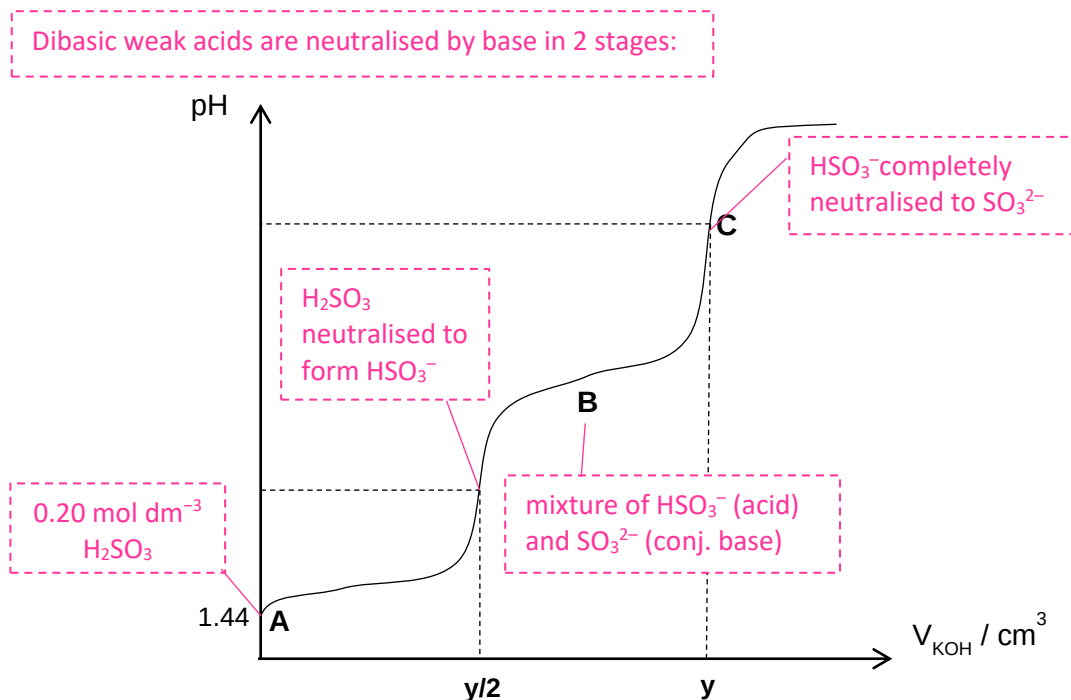


Fig. 2.1

- (a) (i) Determine the concentration of H^+ ions present at point A.

$$\begin{aligned} \text{pH} &= -\lg [\text{H}^+] \\ [\text{H}^+] &= 10^{-1.44} = \underline{0.0363 \text{ mol dm}^{-3}} \quad [1] \end{aligned}$$

[1]

Marker's Comments

- Basic calculation. Ensure you are able to convert pH to $[\text{H}^+]$, and vice versa.

- (ii) Write the expression for the first acid dissociation constant, K_{a1} .

$$K_{a1} = \frac{[\text{H}^+][\text{HSO}_3^-]}{[\text{H}_2\text{SO}_3]} \quad [1]$$

[1]

- (iii) Hence, calculate the value of x , the first acid dissociation constant, K_{a1} , of H_2SO_3 .

When H_2SO_3 dissociates, $[\text{H}^+] = [\text{HSO}_3^-] = 0.0363 \text{ mol dm}^{-3}$

$$K_{a1} = \frac{[\text{H}^+][\text{HSO}_3^-]}{[\text{H}_2\text{SO}_3]}$$

$$= (0.0363)^2 / 0.20$$

$$= \underline{6.59 \times 10^{-3} \text{ mol dm}^{-3}} \text{ [1] – allow ecf from a(i)}$$

[1]

- (b) Point **B** of the titration is made up of a mixture that is able to maintain a fairly constant pH when a small amount of acid is added to it.

- (i) Identify the conjugate acid-base pair that is present at point **B**.

HSO_3^- (acid) and SO_3^{2-} (conjugate base) [1]

[1]

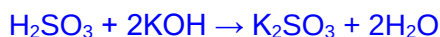
- (ii) Explain, with the aid of an equation, how this mixture at point **B** is able to maintain a fairly constant pH when a small amount of acid is added to it.



The added H^+ is removed as HSO_3^- . $[\text{H}^+]$ is slightly changed hence pH remains fairly constant. [1]

[2]

- (c) (i) Determine the equivalence volume y , at point **C**.



$$\text{Amount of } \text{H}_2\text{SO}_3 = (30/1000) \times 0.20 = \underline{0.006 \text{ mol}} \text{ [1]}$$

Since $1\text{H}_2\text{SO}_3 \equiv 2\text{KOH}$ (for complete neutralisation)

$$\text{Amount of KOH} = 0.006 \times 2 = 0.012 \text{ mol}$$

Equivalence volume, y (i.e. for complete neutralisation)

$$= 0.012 / 0.30 = 0.04 \text{ dm}^3 = \underline{40 \text{ cm}^3} \text{ [1] – } y \text{ is in cm}^3$$

[2]

- (ii) Write an equation to explain why the pH at point **C** is more than 7.



[1]

SO_3^{2-} , being a conjugate base, dissociates in water to produce OH^- ions, resulting in $[\text{OH}^-] > [\text{H}^+]$. Hence, $\text{pH} > 7$.

- (iii) Suggest a suitable indicator that can be used to determine the equivalence point at point **C**.

Phenolphthalein [1]

[1]



- (d) In a separate experiment, a student attempted to analyse the copper content present in a sample of copper ore. First, the copper ore was dissolved in acid to convert all of the copper content into aqueous Cu^{2+} ions. $\text{HSO}_3^-(\text{aq})$ can then act as a reducing agent to reduce the $\text{Cu}^{2+}(\text{aq})$ ions present to form $\text{HSO}_4^-(\text{aq})$ and $\text{Cu}^+(\text{aq})$.

- (i) Determine the change in oxidation state of S from HSO_3^- to HSO_4^- .

The oxidation state changes from +4 (HSO_3^-) to +6 (HSO_4^-) [1]

[1]

- (ii) Construct the half-equation for the oxidation of HSO_3^- to HSO_4^- .



[1]

- (iii) Hence, write the overall equation for the reaction between HSO_3^- and Cu^{2+} .



[1]

Marker's Comments

- part (ii): construction of half-equation is a basic skill. Do revise if unsure.
- part (iii): overall equation should not contain electrons.

- (iv) The $\text{Cu}^+(\text{aq})$ ions can then be determined by precipitation upon the addition of $\text{NaSCN}(\text{aq})$, leading to the formation of the highly insoluble salt $\text{CuSCN}(\text{s})$. A student obtained a 4.05 g of CuSCN precipitate from a 2.40 g of ore sample at the end of the reaction. Calculate the percentage by mass of copper in this sample of ore.



$$\text{Amount of CuSCN} = 4.05 / (63.5 + 32.1 + 12 + 14) = 3.33 \times 10^{-2} \text{ mol} \quad [1]$$

$$\text{Since CuSCN} \equiv \text{Cu}^+ \equiv \text{Cu}^{2+}$$

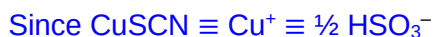
$$\text{Amount of Cu}^{2+} = 3.33 \times 10^{-2} \text{ mol} \quad [1]$$

$$\text{Mass of Cu present} = 3.33 \times 10^{-2} \times 63.5 = 2.11 \text{ g}$$

$$\text{Percentage by mass of copper in ore} = (2.11 / 2.40) \times 100\% = 87.9\% \text{ (3 s.f.)} \quad [1]$$

[3]

- (v) The concentration of the $\text{HSO}_3^-(\text{aq})$ solution used in the experiment was 55.7 g dm^{-3} . Using information from (iv), calculate the volume of $\text{HSO}_3^-(\text{aq})$ solution used in the experiment.



$$\text{Amount of HSO}_3^- = 3.33 \times 10^{-2} / 2 = 1.665 \times 10^{-2} \text{ mol}$$

[1] allow for ecf from d(iii) mole ratio & d(iv)

$$\begin{aligned} \text{Concentration of HSO}_3^- \text{ in mol dm}^{-3} \\ = 55.7 / (1 + 32.1 + 48) = 0.687 \text{ mol dm}^{-3} \end{aligned}$$

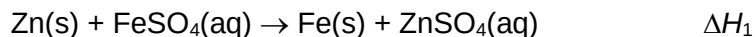
$$\begin{aligned} \text{Volume of HSO}_3^- &= 1.665 \times 10^{-2} / 0.687 \\ &= 2.42 \times 10^{-2} \text{ dm}^3 = 24.2 \text{ cm}^3 \quad [1] \end{aligned}$$

[2]

[Total: 18]

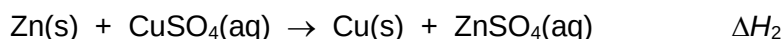


- 3 A student wanted to investigate the enthalpy change of the following reaction.



He went to the laboratory and tried looking for zinc metal and iron(II) sulfate solution to carry out the experiment. Unfortunately, the laboratory did not have any iron(II) sulfate solution. Instead, he managed to find zinc metal, iron metal and copper(II) sulfate solution. He decided to use his knowledge of Hess' Law to perform two experiments to determine ΔH_1 .

- (a) For the first experiment, he reacted excess zinc metal with $0.600 \text{ mol dm}^{-3}$ copper(II) sulfate solution.



He performed the following procedure:

1. Measure 30 cm^3 of $0.600 \text{ mol dm}^{-3}$ copper(II) sulfate solution into a styrofoam cup.
2. Measure the temperature for first 1.5 minutes at every 30 seconds.
3. Add all the zinc metal to the copper(II) sulfate in the styrofoam cup at exactly 2 minutes.
4. Measure the temperature at 2.5 minutes, then every 30 seconds after mixing until the temperature reaches a maximum. Thereafter, measure the temperature every 1 min after the maximum temperature has been reached.

After he recorded the timings and the temperature readings, the student plotted a graph as shown on page 9 to determine the temperature change, ΔT , using a graphical method.



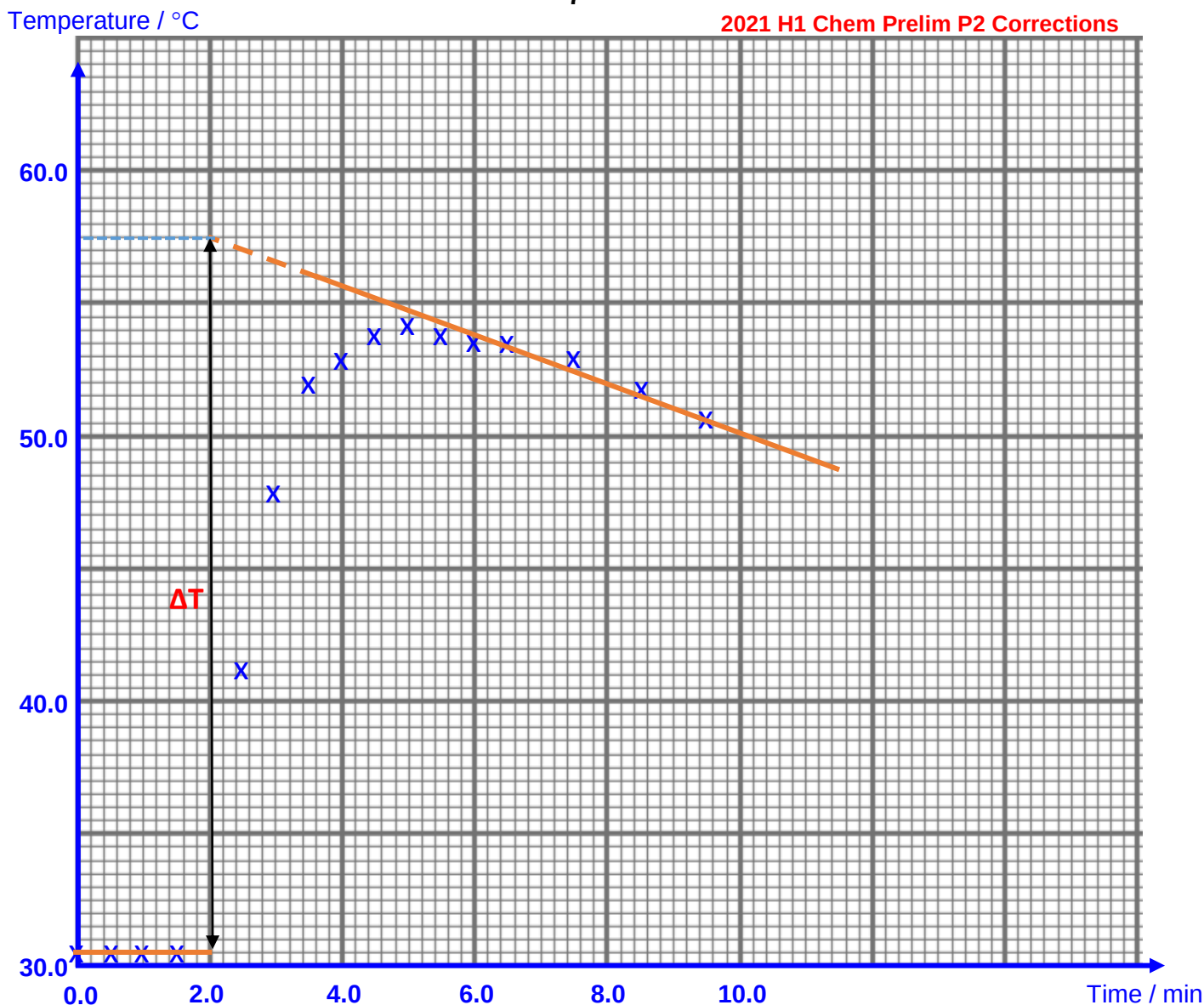


Fig 3.1

- (i) On the graph, draw two best-fit straight lines, the first taking into account the points before zinc was added, and the second taking into account the points after the reaction had finished.

Extrapolate (extend) both lines to two minutes.

Hence, determine the value of temperature change, ΔT , for ΔH_2 .

$$\Delta T = T_{\max} - T_{\text{initial}} = 57.5 - 30.5 = 27.0 \text{ } ^\circ\text{C} \text{ [1]}$$

[1]

Marker's Comments

- Initial temperature, T_{initial} , can read by drawing a best fit straight line through the points before zinc was added, i.e. from 0.0 min to 2.0min. Hence, $T_{\text{initial}} = 30.5^\circ\text{C}$.
- After reaction stops (indicated by highest temperature at ~5.0min), temperature starts to fall as solution cools. Best-fit straight line can be drawn through the points after reaction stops, then extended to the time that zinc was added, i.e. 2.0min. This extension helps to account for heat loss to the surroundings during the experiment. For this line, the temperature read at time = 2.0min gives the maximum temperature, $T_{\max}$.



- (ii) Using your answer from (i), calculate the energy produced during the reaction of zinc with copper(II) sulfate solution.

You should assume that the specific heat capacity of the reaction mixture is $4.18 \text{ J g}^{-1} \text{ K}^{-1}$ and the density of the solution is 1.00 g cm^{-3} .

$$\begin{aligned} Q &= m_{\text{sol}} c_{\text{sol}} \Delta T_{\text{sol}} \\ &= 30 \times 4.18 \times 27.0 \\ &= 3385.8 \text{ J} = \underline{\underline{3390 \text{ J}}} \text{ (3 s.f.) [1] – ecf from on } \Delta T \text{ in (i)} \end{aligned}$$

[1]

- (iii) Calculate the amount of copper(II) sulfate in 30.0 cm^3 of the copper(II) sulfate solution. Hence, calculate the enthalpy change for ΔH_2 , in kJ mol^{-1} .

$$\begin{aligned} \text{Amount of CuSO}_4 &= 0.600 \times 0.030 = \underline{\underline{0.0180 \text{ mol}}} \text{ [1]} \\ \Delta H_2 &= -(3385.8 \div 0.0180) \text{ [1] – method} \\ &= -1.881 \times 10^5 \text{ J mol}^{-1} \\ &= \underline{\underline{-188.1 \text{ kJ mol}^{-1}}} \text{ [1] – sign + ans in kJ mol}^{-1} \end{aligned}$$

[3]

Marker's Comments

- For $q = m_{\text{sol}} c \Delta T$, m_{sol} = mass of solution only, as q = heat absorbed by solution
- For **exothermic** reactions, enthalpy change (ΔH) has **negative sign**.

- (b) For the second experiment, he reacted excess iron metal with 30.0 cm^3 of $0.600 \text{ mol dm}^{-3}$ copper(II) sulfate solution and obtained the following result in the table.

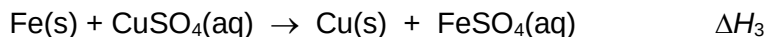


Table 3.2

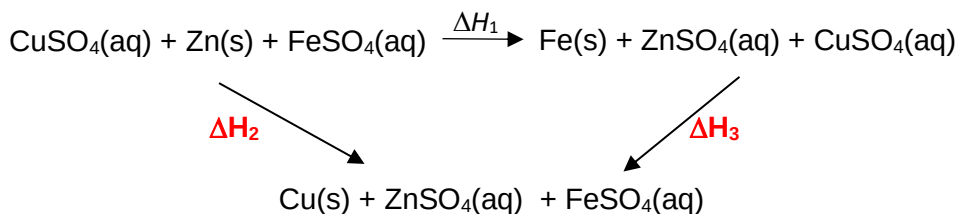
Maximum temperature reached / $^{\circ}\text{C}$	35.2
Initial temperature / $^{\circ}\text{C}$	30.0

Calculate the enthalpy change for ΔH_3 , in kJ mol^{-1} .

$$\begin{aligned} Q &= m_{\text{sol}} c_{\text{sol}} \Delta T_{\text{sol}} = 30 \times 4.18 \times (35.2 - 30.0) = \underline{\underline{652.08 \text{ J}}} \text{ [1]} \\ \Delta H_3 &= 652.08 \div 0.0180 \\ &= -3.62 \times 10^4 \text{ J mol}^{-1} \\ &= \underline{\underline{-36.2 \text{ kJ mol}^{-1}}} \text{ [1] – sign + ans in kJ mol}^{-1} \end{aligned}$$

[2]

- (c) The student constructed the energy cycle below using his knowledge of Hess' Law.



Calculate the enthalpy change for the reaction between zinc and iron (II) sulfate, ΔH_1 .

$$\begin{aligned}
 \Delta H_1 &= \Delta H_2 - \Delta H_3 \\
 &= -188.1 - (-36.2) \text{ [1] - allow ecf} \\
 &= \underline{-151.9 \text{ kJ mol}^{-1}} \text{ [1]}
 \end{aligned}$$

[1]

- (d) The enthalpy change of formation values for ZnSO_4 and FeSO_4 are given in Table 3.3 below.

Table 3.3

compound	$\Delta H_f^\circ / \text{kJ mol}^{-1}$
$\text{ZnSO}_4(\text{s})$	-980
$\text{FeSO}_4(\text{s})$	-929

- (i) Define the term *standard enthalpy change of formation* of $\text{FeSO}_4(\text{s})$.

It is the energy change when one mole of $\text{FeSO}_4(\text{s})$ is formed from its constituent elements, $\text{Fe}(\text{s})$, $\text{S}(\text{s})$ and $\text{O}_2(\text{g})$, under standard conditions (298K and 1 bar). [1]

[1]

Marker's Comments

- Definition has to be **specific** for FeSO_4 . Generic definition is not accepted.

- (ii) Calculate ΔH_4 for the following reaction.



$$\begin{aligned}
 \Delta H_4 &= \Sigma \Delta H_f^\circ(\text{products}) - \Sigma \Delta H_f^\circ(\text{reactants}) \\
 &= \Delta H_f^\circ(\text{ZnSO}_4) - \Delta H_f^\circ(\text{FeSO}_4) \\
 &= -980 - (-929) \text{ [1]} = \underline{-51 \text{ kJ mol}^{-1}} \text{ [1]}
 \end{aligned}$$

[2]

[Total: 11]

Marker's Comments

- Students need to make it a point to remember and use the correct expression when ΔH_f values are given.
- ΔH_f values for elements in standard state = 0

- 4 A reaction scheme involving butan-1-ol is shown in Fig. 4.1.

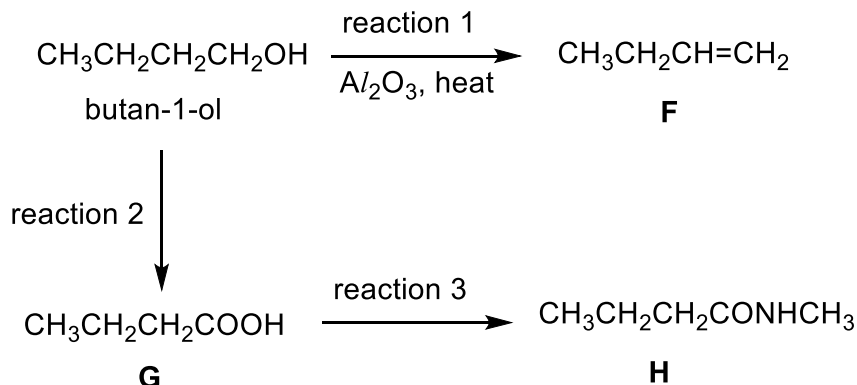


Fig. 4.1

- (a) State the type of reaction for reaction 1.

Elimination [1]

[1]

Marker's Comments

- A vast majority of students correctly identified this reaction.

- (b) State the reagents and conditions required for reaction 2. Write a balanced equation for the reaction that occurs. Use [O] to represent the oxidising agent.

reagents and conditions:

Acidified (OR dilute H_2SO_4), $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$, reflux [1]
 OR Acidified KMnO_4 , reflux

equation:



[2]

Marker's Comments

- A number of students forgot indicate the need for acidic medium.
- Students need to remember that the by-product for this oxidation reaction is H_2O . Note that H_2 gas is formed in acid-metal reactions.

- (c) State the functional group present in compound H, and suggest the reagents and conditions for reaction 3.

functional group:

amide [1]

reagents and conditions:

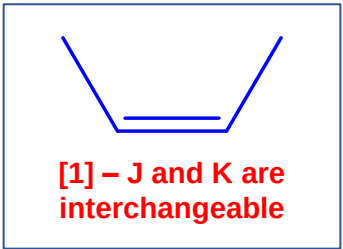
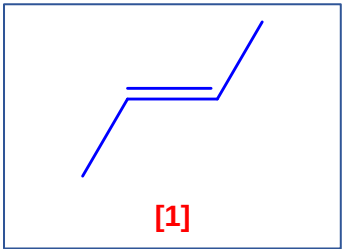
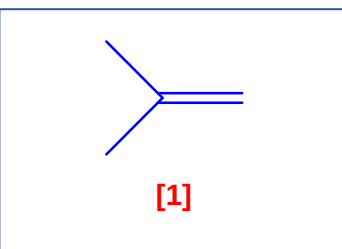
CH_3NH_2 , DCC [1]

[2]

Marker's Comments

- Some students are confused between amines (R-NH_2) and amide (R-CONH-R').
- Many correctly remembered the reaction between carboxylic acids and amines (in presence of DCC) to give amides.
- The specific amine (structural formula or name) needs to be given. Simply stating the functional group is not sufficient.

- (d) Compounds **J**, **K** and **L** are isomers of compound **F**, and contain the alkene functional group. Given that compounds **J** and **K** exist as a pair of cis-trans isomers, suggest the structural formulae for compounds **J**, **K** and **L**.

J	K	L
 [1] – J and K are interchangeable	 [1]	 [1]

[3]

Marker's Comments

- To show cis-trans isomerism, the shape around each C atom of the C=C bond should be trigonal planar.

- (e) Describe a simple chemical test which could be carried out to distinguish between compounds **G** and **H**. In your answer, state clearly the reagents and conditions used in the chemical test and give the expected observations.

To separate samples of **G** and **H**, add solid Na₂CO₃. [1]

Compound G produces effervescence of CO₂ that forms white ppt with limewater while no gas is formed for compound H. [1] – both observations

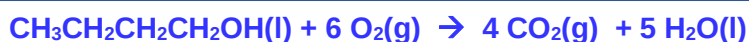
[2]

Marker's Comments

In selecting a suitable chemical test, students should...

- ① **identify the functional group(s)** in the compounds. E.g. G contains carboxylic acid, H contains an amide.
- ② recall possible **reagents** for the respective functional groups that can react to **give a visible observation** (colour change / gas produced / precipitate formed).

- (f) (i) Write an equation to represent the standard enthalpy change of combustion for butan-1-ol, $\Delta H_c^\ominus$.



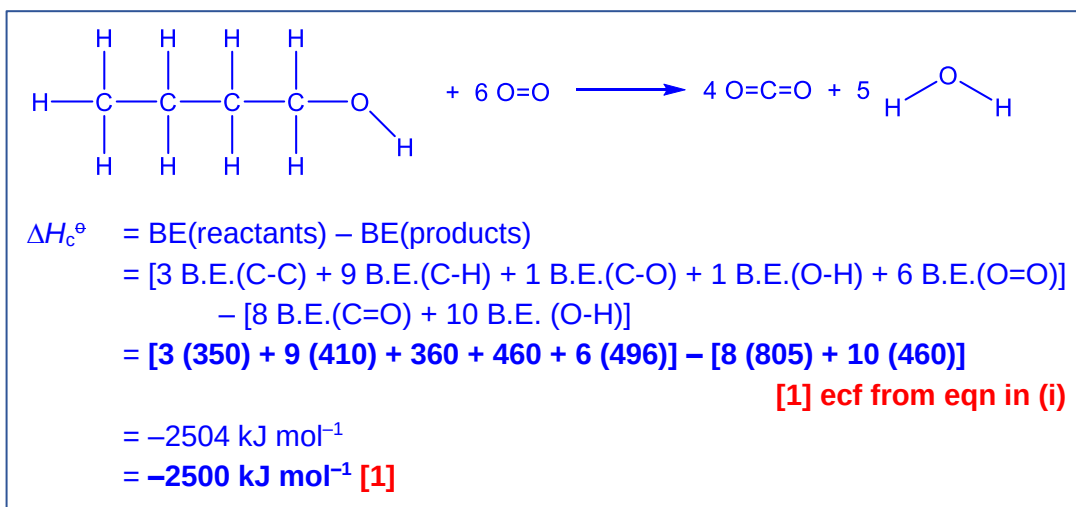
[1] – includes state symbols

[1]

Marker's Comments

- In Chemistry, 'write an equation' generally refers to a **chemical equation**, and not mathematical equation/ expression.
- For enthalpy changes, **state symbols must be included** in the chemical equations.

- (ii) With the use of relevant bond energies from the *Data Booklet*, calculate a value for the enthalpy change of combustion of butan-1-ol.



[2]

Marker's Comments

- Many students are able to correctly determine the number and type of bonds.
- Students need to be meticulous and select the correct bond energy values. Some wrongly selected 740 kJ mol^{-1} instead of 805 kJ mol^{-1} for C=O (in CO_2).

- (iii) The true value of $\Delta H_c^\ominus$ of butan-1-ol is $-2640 \text{ kJ mol}^{-1}$. Suggest a reason for the discrepancy.

The bond energies from the Data Booklet are average values and not be specific to the compounds in the reaction. [1]

OR Bond energies are for breaking bonds in the gaseous state. However, butan-1-ol and water are in the liquid state at standard conditions. [1]

[Total: 14]

Marker's Comments

- Standard question that many were able to correctly answer.

- 5 Some of the most commonly used polymers are formed by the polymerisation of ethenes. The presence of side-chains affects the bulk properties of the polymer. Unbranched polymers pack closer together than polymers with several side-chains.

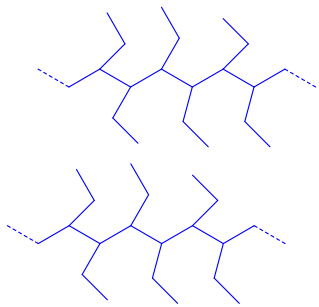
Poly(ethene) exists in two different forms – LDPE (low density poly(ethene)) which contains many side chains, and HDPE (high density poly(ethene)), in which there are fewer and shorter side chains.

- (a) State the type of polymerisation that ethene undergoes in the formation of poly(ethene).

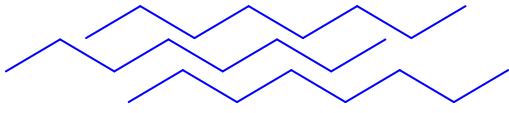
Addition polymerisation [1]

[1]

- (b) Explain, with the aid of suitable diagrams, why the presence of side chains causes a difference in the densities of LDPE and HDPE.



LDPE



HDPE

[1] – minimum of 2 chains for each polymer, with label

In LDPE, the presence of **extensive branching increases the distance between polymer chain molecules** due to **irregular packing**, resulting in a **lower density**. **OR** In HDPE, **little/ no branching** allows **polymer chains to pack closely and decreases the distance between polymer chains**, resulting in **higher density**. [1]

[2]

- (c) By identifying the type of bonding between the poly(ethene) chains, explain why LDPE has a lower melting point than HDPE.

With the extensive branching in LDPE, **less energy** is required to overcome the **weaker instantaneous dipole – induced dipole interactions between polymer chains**. [1]

[1]

Marker's Comments

- Some students mixed up the concepts of LDPE vs HDPE, and thermoplastic vs thermoset. Do revise and ensure ability to distinguish between the 2 sets.

LDPE is classified as a *thermoplastic polymer*, and is widely used in the making of grocery bags and food packaging.

For items that require more rigidity and structure such as vehicle tyres, the use of a *thermosetting polymer* is more appropriate. Thermosetting polymers can be made when polymer chains with specific side chains undergo a chemical process known as curing to produce an infusible and insoluble polymer.

- (d) With reference to their structure, bonding and properties, explain the difference between thermoplastic and thermosetting polymers.

Thermoplastic polymers have weak van der Waals' forces between polymer chains. They are flexible / soften when heated. [1]

Thermosetting polymers have cross-linking/ strong covalent bonds between polymer chains. They are hard and rigid / cannot be softened by heat / decomposes on strong heating. [1]

[2]

Marker's Comments

- Many students are able to clearly distinguish between thermoplastics vs thermosets.

- (e) Poly(ethene) bags pollute the environment for a long time because they cannot be broken down by bacteria and acids in the environment. In recent years, there is increased awareness on the need to reduce consumption of poly(ethene) by using reusable grocery bags in place of single-use ones.

- (i) Explain why poly(ethene) cannot be broken down by acids in the environment.

Poly(ethene) contains **non-polar and strong C-H bonds that are resistant to most chemicals. [1]**

[1]

Marker's Comments

- Explanation should focus on the structure present in poly(ethene) rather than what is absent.

- (ii) Other than reducing the use of poly(ethene), suggest another way to lessen its impact on the environment. Give your reasoning.

Poly(ethene) can be **recycled / reused to reduce the quantity of poly(ethene) to be discarded / produced. [1]**

[1]

[Total: 8]

Marker's Comments

- Students need to answer to the question. The 'reasoning' in this question is to explain how the method you suggested can lessen the impact of poly(ethene) on the environment.



Section B

Answer **one** question from this section in the spaces provided.

- 6 Naturally occurring hydrogen halides are commonly found in places with high volcanic activity. They exist as diatomic gases at room temperature and pressure.

- (a) Some physical properties of three hydrogen halides are given in Table 6.1.

Table 6.1

hydrogen halide	boiling point/ °C	decomposition temperature
HCl	– 85	does not decompose
HBr	– 66	around 2000 °C
HI	– 34	350 °C

- (i) Among the hydrogen halides, HI has the highest boiling point, and yet it has the lowest decomposition temperature. By considering the bonds present, use relevant data from the *Data Booklet* to explain why this is so.

The hydrogen halides have simple molecular structures. **HI** has the largest molecular size, hence the largest electron cloud size, with the greatest extent of distortion of electron cloud. Largest amount of energy is required to overcome the greatest extent of intermolecular instantaneous dipole-induced dipole interaction. Thus, it has the highest melting point.

[2] for 3 points, [1] with 2 points

Whereas thermal decomposition is about breaking the H-X covalent bonds. Since **H-I has the lowest bond energy of 299 kJ mol⁻¹**, compared to that of H-Cl and H-Br, least amount of energy is required to break the H-I bond **[1]**, hence it has the lowest decomposition temperature.

[3]

Marker's Comments

- Students need to know that these are two different processes. The difference in boiling point is due to the extent of intermolecular forces of attraction, while the difference in decomposition temperature is due to the breaking of H-X bonds. Hence students should be clear about what bonds are broken.

- (ii) HCl and MgCl_2 both form acidic solutions when they are dissolved in water.

With the aid of equations, explain the reactions of these chlorides with water. State the expected pH of the resultant solutions formed.

HCl dissolves completely in water to give large amount of H_3O^+ , hence it has a pH of 1 or 2 [1].



MgCl_2 dissolves readily in water to produce Mg^{2+} which causes slight hydrolysis, producing a small amount of H^+ , hence it has a pH of 6.5 [1].

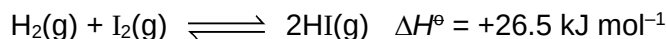


[5]

Marker's Comments

- Some students could not write the dissociation of HCl and state its pH correctly.
- The hydration and slight hydrolysis of MgCl_2 is basic book work which should be properly memorised.

Hydrogen iodide can be synthesised by reacting hydrogen with iodine. The reaction is allowed to reach an equilibrium.



- (b) The equilibrium constant, K_c , for this reaction at 200°C is 46.9.

- (i) 1 mole each of H_2 and I_2 were reacted in a 2 dm^3 vessel at 200°C . Calculate the concentration of HI that could be obtained when equilibrium was reached.

	$\text{H}_2(\text{g})$	$+ \text{I}_2(\text{g})$	$\rightleftharpoons$	$2\text{HI}(\text{g})$
initial amt/ mol	1	1		0
change in amt/ mol	$-x$	$-x$		$+2x$
equilibrium amt/ mol	$1-x$	$1-x$		$2x$

[1]

$$K_c = \frac{[\text{HI}]^2}{[\text{H}_2][\text{I}_2]} \text{ [1] for correct } K_c \text{ expression}$$

$$46.9 = \frac{\left(\frac{2x}{2}\right)^2}{\left(\frac{1-x}{2}\right)^2}$$

$$x = 0.774$$

$$[\text{HI}] \text{ at equilibrium} = \frac{2 \times 0.774}{2} = \underline{0.774 \text{ mol dm}^{-3}} \text{ [1]}$$

[3]



- (ii) With the use of Le Chatelier's Principle, state and explain the effect of increasing total pressure on this equilibrium.

There is no effect on the equilibrium position [1] as there are equal number of moles of gaseous reactants and products [1].

[2]

- (iii) A sketch of the graph of $[HI]$ against time at $200\text{ }^{\circ}\text{C}$ is given in Fig. 6.2.

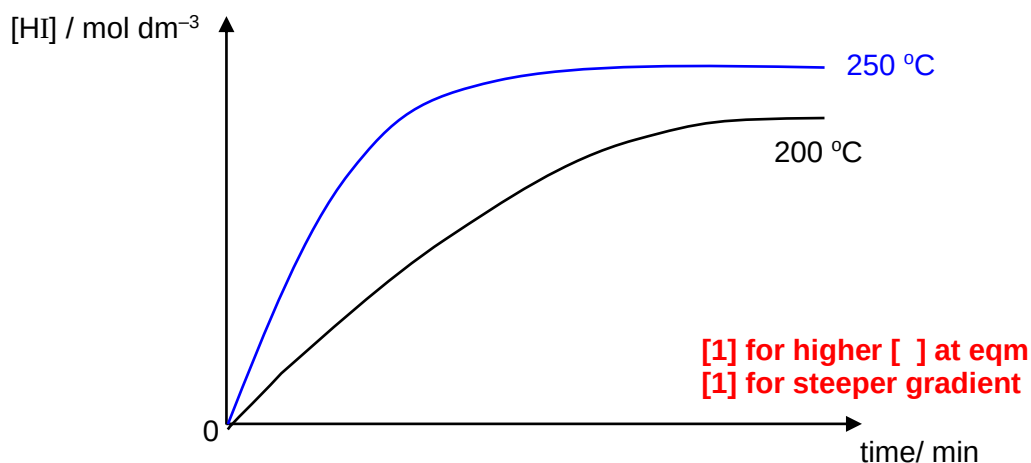


Fig. 6.2

Sketch, on the same axes, the graph if the reaction is conducted at $250\text{ }^{\circ}\text{C}$.

[2]

At higher temperature,

- By Le Chatelier's Principle, equilibrium position will shift right towards the endothermic reaction to absorb heat, hence more HI is produced.
- Rate of reaction will increase, hence gradient becomes steeper.

Marker's Comments

- Most students drew only either one of the above.
- Those who showed a steeper gradient should make it more obvious rather than a gradient almost similar to the original one.

- (c) The rate of reaction of the synthesis of HI is studied by monitoring the concentration of I_2 with time. Two experiments, with $[H_2]$ kept constant at 0.5 mol dm^{-3} and 1.0 mol dm^{-3} are plotted as shown in Fig. 6.3.

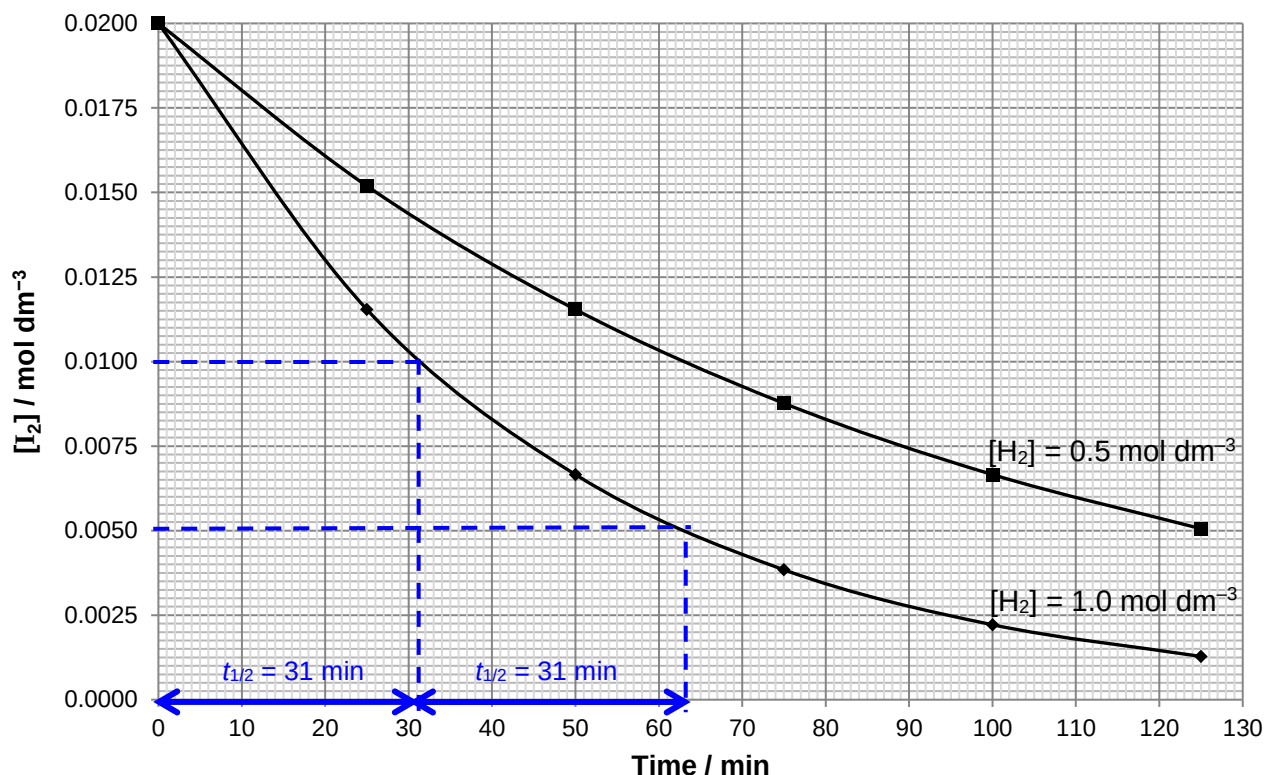


Fig. 6.3

Deduce the order of reaction with respect to I_2 and H_2 .

The half-life is constant at 31 s, thus order of reaction w.r.t. I_2 is 1. [1]

Show clear labelled construction lines to determine half-life on the graph [1]

When $[H_2] = 1.0 \text{ mol dm}^{-3}$,

Initial Rate = | gradient of the graph at $t = 0$ |

$$= \left| \frac{0.0200 - 0.0050}{0 - 30} \right| \quad (\text{gradient needs to be at least half of graph})$$

$$= \underline{5.00 \times 10^{-4} \text{ mol dm}^{-3} \text{ s}^{-1}}$$

When $[H_2] = 0.50 \text{ mol dm}^{-3}$,

Initial Rate = | gradient of the graph at $t = 0$ |

$$= \left| \frac{0.0200 - 0.0100}{0 - 40} \right| \quad (\text{gradient needs to be at least half of graph})$$

$$= \underline{2.50 \times 10^{-4} \text{ mol dm}^{-3} \text{ s}^{-1}} \quad \text{*Rate is always positive}$$

[1] for calculation of gradients

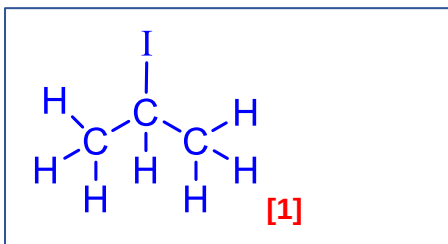
When $[H_2]$ is increased 2 times, while keeping the $[I_2]$ constant, rate of reaction is increased approximately 2 times. Hence, order of reaction wrt H_2 is 1. [1].

[4]



- (d) Hydrogen halides are commonly used in organic synthesis.

Draw the displayed formula of the major product when hydrogen iodide, HI, reacts with propene.



[1]

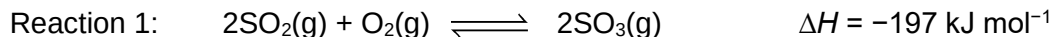
Marker's Comments

- This is the major product formed according to Markovnikov's rule.
- However, most students gave the minor product, 1-iodopropane which is wrong.
- Students should read the question carefully and show the displayed formula.

[Total: 20]

- 7 Sulfur dioxide, SO_2 , is an air pollutant which causes environmental and health problems. Yet it is produced in large quantities every year for the synthesis of sulfuric acid.

The Contact Process is an industrial method for producing sulfuric acid. One of the steps involves the oxidation of SO_2 to produce sulfur trioxide, SO_3 , according to the following equilibrium.



- (a) At room temperature and pressure, SO_2 reacts slowly with oxygen to produce SO_3 . Hence, in the industry, this step is conducted under the following conditions, using vanadium(V) oxide as a catalyst.

pressure 2 atm
 temperature 450 °C
 catalyst V_2O_5

- (i) Suggest the **effect** each of these industrial conditions has on

- the position of equilibrium,
- the rate of the reaction.

Higher pressure

When a **higher pressure** is used (greater than 1 atm or 101 kPa), by Le Chatelier's Principle, the equilibrium position will shift to the right towards a reduction in the number of moles of gas to reduce pressure. [1]

The rate of reaction will be higher. [1]

Higher temperature

When a **higher temperature** (greater than room temperature) is used, by Le Chatelier's Principle, the equilibrium position will shift to the left towards an endothermic reaction so as to absorb heat. [1]

The rate of reaction will be higher. [1]

Catalyst

It does not affect the equilibrium position. The rate of reaction for both forward and backward will be higher. [1]

[5]

Marker's Comments

- The most common mistake is the failure to see that the 2 atm here is at a higher pressure than the room pressure condition, hence equilibrium position will shift because of this rise in pressure.



- (ii) At equilibrium, the concentration of SO_2 is $0.100 \text{ mol dm}^{-3}$, concentration of O_2 is $0.200 \text{ mol dm}^{-3}$ and the concentration of SO_3 is $0.0930 \text{ mol dm}^{-3}$. Write the expression for the equilibrium constant K_c and calculate the value of K_c .

$$K_c = \frac{[\text{SO}_3]^2}{[\text{SO}_2]^2 [\text{O}_2]} \quad [1]$$

$$K_c = \frac{[0.0930]^2}{[0.100]^2 [0.200]} = \underline{4.32} \quad [1] \text{ mol}^{-1} \text{ dm}^3$$

[2]

- (b) The kinetics of reaction 1, the oxidation of SO_2 to produce SO_3 , is studied at constant temperature.

Table 7.1

experiment	initial concentration of SO_2 / mol dm^{-3}	initial concentration of O_2 / mol dm^{-3}	initial rate / $\text{mol dm}^{-3} \text{ s}^{-1}$
1	0.01	0.2	1.61×10^{-3}
2	0.02	0.4	6.42×10^{-3}
3	0.04	0.4	1.28×10^{-2}

- (i) Using the data in Table 7.1, deduce the orders of reaction with respect to SO_2 and O_2 .

Comparing experiment 2 and 3,
When $[\text{O}_2]$ remained constant and $[\text{SO}_2]$ doubled, the initial rate also doubled. Hence order of reaction w.r.t. SO_2 is 1 [1].

Comparing experiment 1 and 2,
When $[\text{SO}_2]$ doubled and $[\text{O}_2]$ also doubled, the initial rate is 4 times the original. Since the order of reaction w.r.t. SO_2 is 1, hence order of reaction w.r.t. O_2 is 1 [1].

OR using calculation method.

$$\frac{\text{rate}_1}{\text{rate}_2} = \frac{[\text{SO}_2][\text{O}_2]^x}{[\text{SO}_2][\text{O}_2]^x}$$

$$\frac{1.61 \times 10^{-3}}{6.42 \times 10^{-3}} = \frac{0.01 \times 0.2^x}{0.02 \times 0.4^x}$$

$x = 1$, $\therefore$ the order of reaction w.r.t. O_2 is 1.

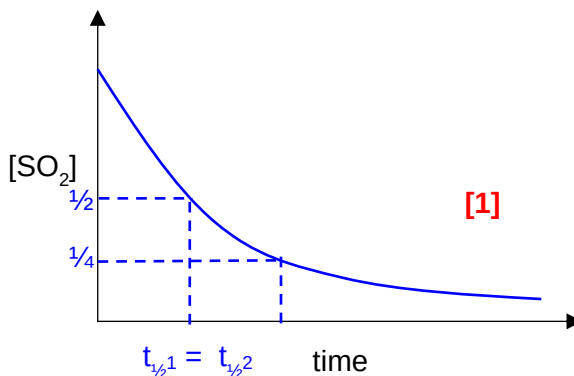
[2]

- (ii) Hence, write the rate equation for this reaction.

$$\text{Rate} = k [\text{SO}_2] [\text{O}_2] \quad [1]$$

[1]

(iii) Complete Fig. 7.2 to show how $[\text{SO}_2]$ varies with time at constant temperature.



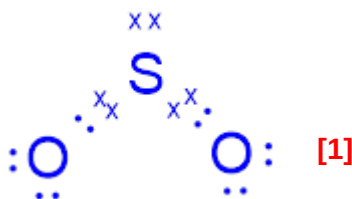
Marker's Comments

- Students should indicate that $t_{1/2}$ is constant to show that it is 1st order w.r.t. SO_2 .

[1]

(c) Although SO_2 is an oxide of a Period 3 element, its chemical and physical properties differ from some of the other oxides, like magnesium oxide, MgO , and aluminium oxide, Al_2O_3 .

(i) Draw a 'dot-and-cross' diagram to show the bonding in SO_2 . Using Valence Shell Electron Pair Repulsion theory, state its shape and bond angle.



Shape: **bent**

Bond angle: **119° [1] for both**

[2]

Marker's Comments

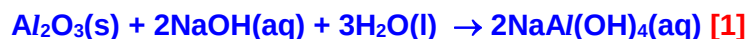
- Not many students got the shape and bond angle correct.
- Students should revise the VSEPR concept properly.

- (ii) Write equations for any reactions that these three oxides would have with hydrochloric acid and with sodium hydroxide. Identify any of these oxides that shows no reaction.

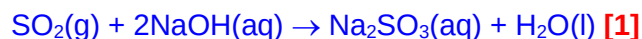
Reaction of MgO with acid



Reaction of Al₂O₃ with acid and base



Reaction of SO₂ with base



MgO shows no reaction with NaOH and SO₂ shows no reaction with HCl [1]

Marker's Comments

- This is a piece of book work which can be scored easily. Students should make it a point to memorise all the reactions and equations.
- Some students could not figure out the product of the reaction of SO₂ with NaOH, and wrote Na₂SO₄, but had the problem of balancing the equation.
- Note:
 - $\text{SO}_3 + 2\text{NaOH} \rightarrow \text{Na}_2\text{SO}_4 + \text{H}_2\text{O}$
 - $\text{SO}_2 + 2\text{NaOH} \rightarrow \text{Na}_2\text{SO}_3 + \text{H}_2\text{O}$

- (iii) Explain, in terms of structure and bonding, why SO₂ has a much lower melting point than MgO.

SO₂ has a simple molecular structure. Less amount of energy is required to overcome the weak intermolecular permanent dipole-permanent dipole interaction [1]. Hence it has a low melting point.

MgO has a giant ionic lattice structures. Large amount of energy is required to overcome the strong electrostatic forces of attraction between the oppositely charged ions [1]. Hence they have much higher melting points

[2]

[Total: 20]

